

schema_version: "0.4"

01 — Business Requirements Specification (BRS)

Phase 1. Translate the business need into a structured, unambiguous requirements document. After this is frozen, the work **forks** into Design Thread (**02a**) and Build Thread (**02b**) — both run in parallel.

Project: GroFast · **ID:** **grofast** · **Lead:** Devendra Sharma

1. Executive summary

GroFast is a hyperlocal quick-commerce grocery delivery platform for the Indian market, modeled on Zepto's operating playbook with a hybrid fulfillment twist: company-run dark stores deliver in 10–20 minutes inside dense urban zones, while partnered local stores extend coverage where dark stores don't yet exist. V1 ships four clients — a customer mobile app, a customer web storefront, a rider app, and an admin + ops dashboard — backed by a single scalable platform covering catalog, ordering, secure payments, real-time order tracking, inventory management, and operational analytics. The goal of V1 is to prove the unit economics and delivery promise in one launch city before multi-city expansion.

2. Business objectives

#	Objective	Success metric	Priority
1	Deliver on the speed promise in dark-store zones	≥ 85% of dark-store orders delivered within the ETA promised at checkout; median delivery ≤ 20 min	P0
2	Establish reliable order fulfillment end-to-end	Order fulfillment rate ≥ 95% (placed → delivered, excl. customer cancellations); item fill rate ≥ 97%	P0
3	Convert browsers to buyers with a frictionless ordering flow	Checkout conversion ≥ 60% from cart; order placement flow ≤ 90 seconds for a returning customer	P0
4	Achieve payment reliability and trust	Payment success rate ≥ 97% on UPI; refunds auto-initiated within 24 h of a failed/cancelled order	P0
5	Keep stock data honest across the hybrid network	OOS-after-order rate ≤ 2% of line items; partner-store stock feed freshness ≤ 15 min	P1
6	Give ops real-time visibility to run the network	Admin dashboard reflects order/inventory state within 30 s; daily automated KPI report	P1
7	Build repeat-purchase behavior	≥ 40% of month-1 customers place a second order within 30 days	P1

3. Stakeholders & user personas

3a. Stakeholders

Name / role	Org function	Interest in this project
Devendra Sharma — Pod Owner / Lead	Product & delivery	Accountable for V1 shipping and lifecycle gates
Ops head (TBD)	Dark-store & fleet operations	Picker/rider tooling, SLA dashboards, zone management
Partner success (TBD)	Marketplace	Partner-store onboarding, stock feed quality, commission settlement
Finance (TBD)	Payments & reconciliation	Payment success, refunds, COD reconciliation, GST invoicing
Engineering lead (TBD)	Build	Architecture, scalability, on-call

3b. User personas

Persona	Who they are	What they need from this product	Frequency of use
Urban convenience shopper ("Priya")	22–40, metro resident, time-poor, UPI-native	Find items fast, order in under 2 minutes, trust the ETA, track the rider live	2–5×/week
Household planner ("Sunita")	Weekly big-basket buyer, price-sensitive	Larger carts, offers/coupons, scheduled-ish reliability via partner stores	1–2×/week
Rider ("Ravi")	Gig delivery partner on a two-wheeler	Clear job offers, optimal pickup/drop navigation, earnings visibility, minimal app friction	Daily, all shift
Dark-store picker ("Amit")	Store staff assembling orders	Fast pick lists sorted by store layout, barcode verification, OOS substitution flow	Daily, all shift
Ops manager ("Neha")	City ops team	Live order/rider/store dashboards, exception queues, inventory controls, zone/ETA config	Daily
Partner-store owner ("Manoj")	Local kirana/supermarket partner	Order notifications, stock/price management, settlement reports	Daily

4. Scope

4a. In scope (V1)

- [x] Customer mobile app (Android + iOS) and responsive customer web storefront
- [x] Rider app (Android) for job assignment, navigation handoff, and delivery completion
- [x] Admin + ops dashboard (web) incl. picker workflow, inventory, catalog, zone, and analytics modules
- [x] Phone-OTP customer auth; address book with geolocation
- [x] Catalog with categories, search, item detail; per-store availability and pricing
- [x] Cart, checkout, ETA promise, delivery fees, coupons (flat/percent)
- [x] Payments: UPI, cards, wallets via PCI-compliant gateway; COD (pending sign-off — see §13); refunds
- [x] Order lifecycle: placed → packed → dispatched → delivered / cancelled, with real-time customer tracking
- [x] Rider dispatch: auto-assignment to nearest available rider, manual override by ops
- [x] Inventory management: dark-store stock CRUD + adjustments + low-stock alerts; partner-store stock ingestion via API/CSV
- [x] Serviceability zones (geofenced polygons) with store-to-zone mapping and dynamic ETA computation
- [x] Notifications: push + SMS for order events; OTP delivery
- [x] Analytics: ops KPIs (orders, SLA, fill rate, rider utilization), sales reports, daily digest
- [x] Single launch city — **Delhi NCR** (decided 2026-07-10) — single currency (INR), English UI

4b. Out of scope (V1)

- [] Ads & sponsored listings (→ FU-1)
- [] Subscriptions / memberships (e.g., daily milk, free-delivery pass) (→ FU-2)
- [] Loyalty points & referral program (→ FU-3)
- [] Scheduled/slot delivery (→ FU-4)
- [] Multi-city, multi-language, multi-currency (→ FU-5)
- [] In-house rider payroll/incentive engine (basic earnings display only) (→ FU-6)
- [] Warehouse-to-dark-store supply chain / procurement (inventory arrives via manual GRN in V1) (→ FU-7)
- [] Customer support chat/CRM integration (phone + email only in V1) (→ FU-8)

4c. Assumptions

- [x] Launch city: Delhi NCR (initial zone polygons to be drawn in ops config before launch)
- [] Payment gateway will be a PCI-DSS Level 1 Indian provider (Razorpay-class); GroFast never stores raw card data
- [] Riders are onboarded/KYC'd through an ops-managed back-office process (not self-serve in V1)
- [] Partner stores can provide stock via a simple API or scheduled CSV; deep POS integrations are per-partner projects post-V1
- [] Cloud: managed Kubernetes-capable public cloud (AWS-class); final vendor pending (§13)

5. Functional requirements

Stable IDs. **Never renumber** — strike through with ~~FR-x~~ if removed.

ID	Title	User story	Acceptance criteria	Priority
FR-1	Phone-OTP signup/login	As a customer, I want to log in with my phone number and OTP, so that I don't manage passwords	Given a valid Indian mobile number, when I request an OTP, then I receive it within 30 s and can log in with ≤ 2 attempts; sessions persist via refresh tokens	P0
FR-2	Address book & location detection	As a customer, I want to save addresses and auto-detect my location, so that serviceability and ETA are accurate	Given GPS permission, when I open the app, then my deliverable address is resolved and the serving store is selected; addresses support label, floor/landmark	P0
FR-3	Serviceability check	As a customer, I want to know immediately if my location is serviceable, so that I don't build a dead cart	Given a location outside all zones, when I browse, then a "not serviceable yet" state with notify-me capture is shown	P0
FR-4	Browse catalog by category	As a customer, I want to browse categorized products, so that I can shop without searching	Given the serving store, when I open a category, then only in-stock-priced items for that store render, p95 < 200 ms server time	P0
FR-5	Product search with typo tolerance	As a customer, I want fast, forgiving search, so that I find items even with misspellings	Given "atta", "aata", or "attaa", when I search, then relevant flours rank in top results; results return < 300 ms p95; empty results suggest alternates	P0
FR-6	Product detail page	As a customer, I want item details (images, weight, MRP vs price, expiry-sensitive info), so that I can decide	Given any SKU, when I open detail, then price, discount, pack size, and availability at my serving store are correct	P1
FR-7	Cart management	As a customer, I want to add/remove/step quantities with a running total, so that I control my order	Given stock limits, when I exceed available qty or per-order caps, then the cart blocks the increment with a reason; cart persists across sessions and devices	P0
FR-8	ETA promise at checkout	As a customer, I want a committed delivery estimate before paying, so that I can trust the promise	Given store load and rider availability, when I reach checkout, then a dynamic ETA (e.g., "12 min") displays and is stored on the order for SLA measurement	P0
FR-9	Fees, coupons & bill summary	As a customer, I want an itemized bill with delivery/small-cart fees and coupon application, so that pricing is transparent	Given an eligible coupon, when applied, then the discount reflects instantly and invalid coupons show the failure reason; GST-compliant invoice generated on delivery	P0
FR-10	Online payment		Given gateway success, when payment completes, then order confirms in < 3 s; given failure/	P0

ID	Title	User story	Acceptance criteria	Priority
		As a customer, I want to pay via UPI/cards/wallets, so that checkout is instant	timeout, then retry or switch method without losing the cart; webhook reconciliation handles dropped callbacks	
FR-11	Cash on delivery	As a customer, I want COD as a fallback, so that I can order without digital payment	Given COD enabled for my order value, when I place the order, then the rider app shows amount to collect and marks collection at delivery (pending §13 sign-off)	P1
FR-12	Order placement & stock reservation	As a customer, I want my order accepted only if items are really available, so that I'm not disappointed later	Given checkout, when I pay, then stock is atomically reserved for all line items or the order fails cleanly with refund auto-initiated	P0
FR-13	Real-time order tracking	As a customer, I want live status and rider location on a map, so that I know when to be at the door	Given a dispatched order, when I open tracking, then rider position updates $\leq$ every 5 s and status transitions push instantly	P0
FR-14	Order cancellation & refunds	As a customer, I want to cancel before dispatch and get automatic refunds, so that mistakes aren't costly	Given an order not yet dispatched, when I cancel, then reserved stock releases and prepaid amounts auto-refund within 24 h	P0
FR-15	Order history & reorder	As a customer, I want past orders and one-tap reorder, so that repeat shopping is instant	Given a past order, when I tap reorder, then available items land in cart and unavailable ones are flagged	P1
FR-16	Ratings & delivery feedback	As a customer, I want to rate the order and delivery, so that quality issues surface	Given a delivered order, when I rate $\leq$ 3 stars, then a reason picker appears and the record lands in the ops exception queue	P2
FR-17	Picker pick-list workflow	As a picker, I want an optimized pick list per order, so that I can pack fast	Given a new dark-store order, when I accept it, then items list in store-layout order with barcode-scan verification and OOS substitution/refund flow; pack completion timestamps recorded	P0
FR-18	Rider job offer & assignment	As a rider, I want nearby delivery jobs offered automatically, so that I earn without hunting	Given an order packed (or partner order ready), when dispatch runs, then the nearest eligible online rider gets a 30 s accept/decline offer; declines cascade to next rider; ops can force-assign	P0
FR-19	Rider pickup & delivery flow	As a rider, I want clear pickup/drop steps with navigation handoff, so that deliveries are error-free	Given an accepted job, when I proceed, then OTP/scan verification at pickup and delivery-proof (OTP or photo) at drop are enforced; COD amount shown when applicable	P0

ID	Title	User story	Acceptance criteria	Priority
FR-20	Rider availability & earnings view	As a rider, I want to toggle online/offline and see my day's deliveries and earnings, so that I control my shift	Given completed deliveries, when I open earnings, then per-order payout and daily totals are accurate	P1
FR-21	Catalog & SKU management (admin)	As an ops/catalog manager, I want to CRUD products, categories, images, and per-store price/availability, so that the storefront stays correct	Given a catalog change, when saved, then storefronts reflect it within 60 s; bulk CSV upload supported	P0
FR-22	Dark-store inventory management	As an ops manager, I want stock receiving (GRN), adjustments, and cycle counts with audit trail, so that system stock matches shelf stock	Given any stock mutation, when committed, then an immutable audit record (who/when/why) exists; low-stock alerts fire at configurable thresholds	P0
FR-23	Partner-store onboarding & stock sync	As a partner manager, I want to onboard partner stores and ingest their stock/prices via API or CSV, so that marketplace coverage grows	Given a partner feed, when ingested, then SKUs map/merge to the master catalog with a rejection report; feeds older than 15 min flag the store as stale (degraded listing)	P0
FR-24	Zone & ETA configuration	As an ops manager, I want to draw serviceability polygons, map stores to zones, and tune ETA parameters, so that promises match capacity	Given a zone edit, when published, then serviceability and store selection update for new sessions within 60 s	P0
FR-25	Live ops dashboard & exception queue	As an ops manager, I want live order/rider/store views and an exception queue (delayed, stuck, failed-payment, low-rated), so that I intervene fast	Given an order breaching its promised ETA, when the breach occurs, then it appears in the exception queue within 30 s with one-click actions (reassign rider, call customer, cancel-refund)	P0
FR-26	Order management (admin)	As an ops agent, I want to search any order and act on it (modify status, refund, cancel), so that support issues resolve	Given an order ID or phone number, when searched, then full order timeline with every state transition and actor renders	P0
FR-27	Analytics & reports	As a stakeholder, I want KPI dashboards (GMV, orders, AOV, SLA hit rate, fill rate, rider utilization, zone heatmaps) and CSV export, so that decisions are data-driven	Given a date range, when I load a report, then numbers reconcile with the order ledger; daily digest email/Slack at 8 am	P1
FR-28	Notifications engine	As the platform, I want templated push/SMS events on order milestones, so that customers and riders stay informed	Given each lifecycle transition, when it commits, then the mapped notification dispatches within 10 s with delivery-failure fallback (push → SMS)	P0

ID	Title	User story	Acceptance criteria	Priority
FR-29	RBAC for admin	As an admin owner, I want role-based access (ops, catalog, finance, partner, read-only), so that staff see only their tools	Given a finance-role user, when they open catalog editing, then access is denied and the attempt is audit-logged	P0
FR-30	Substitution & partial fulfillment	As a customer, I want sensible handling when an item is OOS after ordering, so that the rest of my order still arrives	Given an OOS at pick time, when the picker flags it, then the customer gets an instant substitute-or-refund prompt (60 s timeout → auto-refund of that line)	P1

6. Non-functional requirements

ID	Category	Requirement	Target	Priority
NFR-1	Performance	Catalog/search/browse server latency	p95 < 200 ms (browse), < 300 ms (search)	P0
NFR-2	Performance	Order placement end-to-end (pay-confirm → order confirmed)	p95 < 3 s	P0
NFR-3	Availability	Order path (browse → checkout → track) uptime	99.9% monthly; graceful degradation of non-critical modules	P0
NFR-4	Scalability	Horizontal scale without redesign	1 city / 60 stores / 10k orders/day at launch → 10× via scale-out only	P0
NFR-5	Security	Auth, transport, and data at rest	TLS everywhere; JWT short-lived + refresh rotation; AES-256 at rest for PII; no raw card storage (gateway tokenization); OWASP ASVS L2	P0
NFR-6	Compliance	India DPDP Act, GST, PCI (via gateway)	DPDP consent + data-deletion flow; GST-compliant invoices; annual gateway compliance attestation	P0
NFR-7	Freshness	Real-time tracking + inventory propagation	Rider GPS ≤ 5 s; stock changes visible ≤ 60 s; partner feeds ≤ 15 min	P0
NFR-8	Observability	Logging, metrics, tracing, alerting	Structured logs, distributed traces on order path, SLO burn alerts; on-call runbooks	P1
NFR-9	Mobile quality	App size, cold start, offline grace	Cold start < 3 s on mid-range Android; cart/catalog cache survives brief offline	P1
NFR-10	Auditability	Money- and stock-touching mutations	100% covered by immutable audit log with actor + reason	P0

7. Business rules

ID	Rule	Source
BR-1	An order is served by exactly one store (dark or partner); no cross-store carts in V1	Ops model
BR-2	ETA promise shown at checkout is stored on the order and is the SLA baseline for objective #1	Objective 1
BR-3	Stock is reserved at payment-confirmation, released on cancellation/expiry (15 min unpaid hold max)	FR-12
BR-4	Refunds for prepaid orders auto-initiate within 24 h; no manual approval below ₹2,000	Objective 4
BR-5	COD is capped per order (default ₹1,500, ops-configurable) and disabled for first-time users flagged risky	Finance
BR-6	Partner stores with stock feeds staler than 15 min are listed as "delivery may take longer" and excluded from fast-ETA promises	Objective 5
BR-7	Riders may carry max 1 active order in V1 (no batching)	Ops simplicity
BR-8	Delivery-proof (OTP or photo) is mandatory to complete an order; COD additionally requires collection confirmation	Fraud control
BR-9	Age-restricted or regulated items (if listed) require category-level flags and are excluded from V1 catalog	Compliance
BR-10	All prices include GST; invoice issued in the operating entity's GSTIN at delivery	Finance

8. Data flow (high-level)

Customer clients talk to the platform API. A serviceability service resolves location → zone → serving store. Catalog/search reads come from a read-optimized store (cache + search index) fed by the catalog/inventory system of record. Checkout orchestrates: price/coupon validation → ETA computation → payment (gateway redirect/intent + webhook reconciliation) → atomic stock reservation → order creation. The order then flows through an event-driven lifecycle: dark-store orders enter the picker queue; partner orders notify the partner store. On "packed/ready", the dispatch service offers the job to riders; rider GPS streams through a location channel to both ops dashboards and the customer's tracking view. Every lifecycle event fans out to the notifications engine, the analytics pipeline (event log → warehouse → dashboards), and the audit log. Admin dashboard performs privileged reads/writes against the same services under RBAC.

Detailed component architecture, data stores, and protocols land in [02b_architecture.md](#).

9. UI requirements overview

- [x] Speed-first: skeleton loads, optimistic UI on cart ops, sub-100 ms perceived interactions
- [x] Mobile-first design system shared across customer app and web; rider app optimized for one-handed, sunlight-readable use

- [x] Live tracking map as the emotional centerpiece of the post-order experience
- [x] Transparent billing (no hidden fees) and honest stock states (no ghost inventory)
- [x] Admin dashboard: dense, keyboard-friendly, exception-queue-driven (manage by exception, not by watching)
- [x] Accessibility: WCAG 2.1 AA for web surfaces; dynamic type support in apps

10. Dependencies

- [] Payment gateway vendor selection & merchant onboarding (blocker for FR-10; owner: Finance/Lead)
- [] Maps/geo provider selection incl. pricing at tracking volumes (FR-2/13/24)
- [] SMS/WhatsApp DLT registration for OTP + transactional templates (India requirement; 2–4 week lead time)
- [] Cloud account, org, and Kubernetes/managed-services baseline
- [] Partner-store pilot cohort (≥ 10 stores) committed for launch city
- [] Rider supply plan for launch zones (hiring/onboarding owned by Ops)
- [] FSSAI registration & GST setup for operating entity

11. Success criteria

- [] All P0 FRs shipped and passing their acceptance criteria in production
- [] 30-day post-launch: ETA hit rate $\geq 85\%$, payment success $\geq 97\%$, fulfillment $\geq 95\%$, OOS-after-order $\leq 2\%$
- [] Order path availability $\geq 99.9\%$ over first 30 days; zero Sev-1 payment/ledger incidents
- [] $\geq 40\%$ month-1 repeat-order rate
- [] Ops runs a full day purely from the dashboard + exception queue (no DB/engineering intervention)

12. Risks & mitigations

#	Risk	Likelihood	Impact	Mitigation	Owner
1	Partner-store stock feeds are stale/wrong → OOS-after-order spikes	H	H	BR-6 degraded listing, 15-min freshness SLA, fill-rate scorecards, delist repeat offenders	Partner success
2	Rider supply shortfall at peak → ETA promise broken	M	H	Dynamic ETA widening with load, zone-level order throttling, ops force-assign, surge planning	Ops
3	Payment webhook loss/duplication → stuck or double-charged orders	M	H	Idempotent order/payment ledger, reconciliation job, auto-refund SLA (BR-4)	Eng
4	ETA model too optimistic at launch (no historical data)	H	M	Conservative launch ETAs, weekly recalibration, promise-vs-actual dashboard	Eng + Ops
5	DPDP/DLT/GST compliance gaps delay launch	M	H	§10 dependencies started in parallel with build; compliance checklist gate in Phase 8	Lead
6	Scope creep on 4 clients strains V1 timeline	H	M	Hard gates of this lifecycle; §14 backlog discipline; rider app kept minimal (FR-18/19/20 only)	Lead
7	COD fraud/leakage	M	M	BR-5 caps, delivery-proof + collection confirmation (BR-8), COD reconciliation report	Finance

13. Open items

- [x] ~~Launch city~~ — **resolved 2026-07-10: Delhi NCR**; initial zone polygons remain an ops-config task pre-launch
- [] **COD in V1: confirm yes/no** — owner: Devendra Sharma + Finance; default = yes with BR-5 caps; affects FR-11/19, rider flows; decide before `/tasks-freeze`
- [] **Payment gateway vendor** — owner: Lead/Finance; needed before `02b_architecture` freeze
- [] **Cloud vendor** — owner: Eng lead; needed before `02b_architecture` freeze
- [] **Pod composition / named owners for Ops, Finance, Partner success, Eng lead** — owner: Devendra Sharma

14. Future updates backlog

Status values: `deferred` (default) · `promoted` (became an FR/NFR in vN+1) · `dropped` (explicitly killed) · `evergreen` (recurring nice-to-have)

Type values: `FR` (functional) · `NFR` (non-functional) · `BR` (business rule)

ID	Title	Type	Rationale (why not now)	Origin	Priority hint	Status	Promoted to
FU-1	Ads & sponsored listings	FR	Needs traffic scale first; monetization phase 2	§4b	P2	deferred	—
FU-2	Subscriptions / free-delivery membership	FR	Retention lever after PMF signal	§4b	P2	deferred	—
FU-3	Loyalty points & referrals	FR	Growth lever post-launch	§4b	P2	deferred	—
FU-4	Scheduled / slot delivery	FR	Different ops flow; conflicts with speed focus in V1	§4b	P2	deferred	—
FU-5	Multi-city, multi-language	FR	V1 proves one city; NFR-4 keeps the door open	§4b	P1	deferred	—
FU-6	Rider incentive & payout engine	FR	Manual payouts acceptable at launch volume	§4b	P2	deferred	—
FU-7	Procurement & supply-chain (PO → GRN automation)	FR	Manual GRN sufficient for ≤ 10 dark stores	§4b	P1	deferred	—
FU-8	Support chat & CRM integration	FR	Phone/email support adequate at launch volume	§4b	P2	deferred	—
FU-9	Order batching for riders (2+ orders/trip)	FR	Efficiency lever once density exists; BR-7 relaxation	BR-7	P1	deferred	—
FU-10	ML-based demand forecasting & auto-replenishment	FR	Needs sales history to train on	§8	P2	deferred	—

Use `/backlog-promote --id FU-X` for ad-hoc promotion. Otherwise `/project-version-bump` walks the list interactively at next-version planning time.

Revision history

#	Date	Triggered by	Scope of change	Re-frozen by
v1	(initial freeze date)	Initial	(initial freeze)	(filled at first /X-freeze)

Each subsequent /X-freeze appends a row here, tagged with the active CR ID (if any). The most-recent freeze is the current canonical state. The history is the audit trail.

Sign-off

- **Reviewer:**
- **Role:** Business owner + Product Manager
- **Timestamp:**
- **Status:** ☐ Not frozen
- **Notes:**

The `/brs-freeze` skill fills this block, generates `_readable/01_brs.pdf`, and unlocks `02a_screens` + `02b_architecture` in parallel.